Table 1: Comparison of MUST and MUSTARD under timing regimes taken from real systems (all times in ms, one tick = 1 ms; YARN in 1 s ticks). Networks: [A] same-region cross-AZ U=1 (Azure AZ round trip below about 2 ms), [B] continental U=65 (etcd tuning guide, 130 ms round trip), [C] intercontinental U=200 (etcd, 350 to 400 ms), [D] degraded U=2 500 (etcd election cap), [E] partition-delayed U=90 000 (GitHub 2012); suffix p: L is half the median. Times are seconds on one core, medians of three runs unless marked † (fewer runs) or $^{\ddagger n}$ (run with n workers; wall time not comparable); $> t$: not finished after t seconds. *Exec.* counts all complete execution graphs including blocked ones and those with a failed assertion; *Viol.* counts the graphs with at least one failed assertion, once per graph. Configuration columns: lease: clients, rounds, TTL, client pause (b is the boundary $TTL - 2(U - L) - sd$); SWIM: probe period and suspicion timeout; BRP: L=0.5 is the half-unit grid of the transit interval $(0, T_D]$; YARN: ticks of the 1 s NM heartbeat.

	Timing			MUST			MUSTARD		
Configuration	L	U	sd	Exec.	Viol.	Time	Exec.	Viol.	Time
Lease lock (C clients, R rounds, TTL 12 s, client pause p)									
A: C=3, R=3, p=11 997 (b−1)	0	1	0	49 206 872	11 603 400	7 627†	5 680 040	0	8 139†
A: C=3, R=3, p=11 998 (b)	0	1	0	49 206 872	11 603 400	7 627†	6 068 016	330 744	8 711†
A: C=4, R=2, p=11 997 (b−1)	0	1	0	99 628 848	37 607 328	14 628†	11 122 128	0	14 243†
A: C=4, R=2, p=11 998 (b)	0	1	0	99 628 848	37 607 328	14 628†	11 787 624	634 752	15 075†
A: C=6, R=1, p=11 997 (b−1)	0	1	0	85 157 536	51 560 640	11 258†	13 504 576	0	12 513†
A: C=6, R=1, p=11 998 (b)	0	1	0	85 157 536	51 560 640	11 258†	14 142 136	637 560	12 974†
A: C=2, R=6, p=11 997 (b−1)	0	1	0	204 073 324	20 007 188	3 334$^{\ddagger 16}$	22 802 036	0	3 049$^{\ddagger 16}$
A: C=2, R=6, p=11 998 (b)	0	1	0	204 073 324	20 007 188	3 334$^{\ddagger 16}$	24 589 098	1 108 432	6 650$^{\ddagger 8}$
C: C=2, R=5, p=11 599 (b−1)	0	200	0	11 337 388	1 111 508	1 756	1 956 698	0	2 901
C: C=2, R=5, p=11 600 (b)	0	200	0	11 337 388	1 111 508	1 756	2 049 066	92 368	3 008
D: C=2, R=5, p=6 999 (b−1)	0	2 500	0	11 337 388	1 111 508	1 756	1 956 698	0	3 000†
D: C=2, R=5, p=7 000 (b)	0	2 500	0	11 337 388	1 111 508	1 756	2 049 066	92 368	3 144†
D: C=2, R=6, p=6 999 (b−1)	0	2 500	0	204 073 324	20 007 188	3 334$^{\ddagger 16}$	23 480 666	0	5 560$^{\ddagger 16}$
D: C=2, R=6, p=7 000 (b)	0	2 500	0	204 073 324	20 007 188	3 334$^{\ddagger 16}$	–	–	–
A: C=3, R=3, p=11 996 (b−1)	0	1	1	49 206 872	11 603 400	7 627†	5 737 272	0	8 000†
A: C=3, R=3, p=11 997 (b)	0	1	1	49 206 872	11 603 400	7 627†	6 068 016	330 744	8 522†
A: C=2, R=5, p=11 996 (b−1)	0	1	1	11 337 388	1 111 508	1 763	1 956 698	0	2 861†
A: C=2, R=5, p=11 997 (b)	0	1	1	11 337 388	1 111 508	1 763	2 049 066	92 368	3 066†
D: C=2, R=5, p=4 499 (b−1)	0	2 500	2 500	11 337 388	1 111 508	1 763	10 633 430	0	14 373†
D: C=2, R=5, p=4 500 (b)	0	2 500	2 500	11 337 388	1 111 508	1 763	11 047 158	413 728	15 230†
A: C=2, R=5, TTL 60 s, p=59 998 (b)	0	1	0	11 337 388	1 111 508	1 763	2 049 066	92 368	3 017
E: C=2, R=3, p=200	0	90 000	0	34 972	3 428	3.8	37 556	3 580	21
Lease lock with fencing (C clients, R rounds, TTL 12 s, pause p)									
A: C=2, R=5, p=11 998 (b)	0	1	0	94 406 884	0	13 660†	6 461 828	0	9 471†
A: C=5, R=1, p=11 998 (b)	0	1	0	13 012 592	0	1 356	577 328	0	415
D: C=5, R=1, p=7 000 (b)	0	2 500	0	13 012 592	0	1 356	577 328	0	411
SWIM (N nodes, R periods, probe, suspicion)									
lan: N=3, R=7, lan (500, 4 000)	0	1	0	633 567 018	603 567 018	6 129$^{\ddagger 16}$	39 478 033	36 732 890	1 850$^{\ddagger 16}$
lan: N=11, R=1, lan (500, 4 000)	0	1	0	37 939 862	37 290 312	5 553†	23 320 437	22 769 302	12 600†
lan: N=6, R=2, lan (500, 4 000)	0	1	0	14 228 178	13 870 962	1 572	5 442 270	5 247 432	2 514†
wan: N=6, R=2, wan (3 000, 30 000)	0	200	0	14 228 178	13 870 962	1 572	5 442 270	5 247 432	2 516
local: N=6, R=2, local (200, 3 000)	0	2 500	0	14 228 178	13 870 962	1 572	21 953 691	20 824 095	8 440†
local: N=3, R=5, local (200, 3 000)	0	2 500	0	3 517 554	3 217 554	297†	6 073 335	4 647 222	2 708†
NRP (variant, R, M, H, P)									
A: nrp, R=4, M=3, H=1 000, P=2	0	1	0	118 424 493	3 538 558	9 251†	1 492 721	33 704	1 296
A: noopt, R=4, M=3, H=1 000, P=2	0	1	0	314 788 770	31 083 976	2 827$^{\ddagger 14}$	2 502 926	221 668	158$^{\ddagger 14}$
A: leasing, R=4, M=3, H=1 000, P=2	0	1	0	279 661 585	0	3 192$^{\ddagger 8}$	2 257 525	0	236$^{\ddagger 8}$

Configuration	Timing			Must			Mustard		
	L	U	sd	Exec.	Viol.	Time	Exec.	Viol.	Time
D: noopt, R=4, M=3, H=10 000, P=5 000	0	2 500	0	314 788 770	31 083 976	2 827$^{\ddagger 14}$	6 484 980	419 466	459$^{\ddagger 14}$
D: nrp, R=4, M=3, H=10 000, P=5 000	0	2 500	0	118 424 493	3 538 558	9 251†	4 243 153	64 314	3 721
C: nrp, R=4, M=3, H=1 000, P=400	0	200	0	118 424 493	3 538 558	9 251†	3 918 481	64 314	3 312
D: leasing, R=4, M=3, H=10 000, P=5 000	0	2 500	0	279 661 585	0	3 192$^{\ddagger 8}$	6 019 470	0	680$^{\ddagger 8}$
A: nrp, R=5, M=3, H=1 000, P=2	0	1	0	–	–	–	125 536 980	4 896 854	10 460$^{\ddagger 16}$
Kafka/nisshi (T transactions, R retries, W, B)									
A: T=16, R=2, W=30 000, B=100	0	1	0	3 289 304	168 880	1 145	71 192	0	244
A: T=32, R=2, W=30 000, B=100	0	1	0	60 291 240	2 123 872	2 747$^{\ddagger 16}$	504 104	0	393$^{\ddagger 16}$
A: T=8, R=4, W=30 000, B=100	0	1	0	–	–	$>$ 13 478	759 286	0	1 629†
A: T=2, R=5, W=30 000, B=100	0	1	0	58 985 415	1 977 470	11 068†	29 348	0	28†
C: T=32, R=2, W=30 000, B=100	0	200	0	60 291 240	2 123 872	2 747$^{\ddagger 16}$	504 104	0	5 505†
D: T=32, R=2, W=30 000, B=100	0	2 500	0	60 291 240	2 123 872	2 747$^{\ddagger 16}$	504 104	0	5 392†
A: T=2, R=8, W=30 000, B=100	0	1	0	–	–	–	1 427 024	0	1 954†
A: T=2, R=10, W=30 000, B=100	0	1	0	–	–	–	17 537 560	0	4 843$^{\ddagger 16}$
E: T=6, R=3, W=30 000, B=100	0	90 000	0	4 007 641	288 014	924	2 416 521	131 182	3 917†
E: T=8, R=2, W=30 000, B=100 fix	0	90 000	0	734 560	0	177	734 560	0	1 350
Alternating bit (K messages, R retries, W)									
A: K=4, R=2, W=200	0	1	0	6 315 296	0	366	8 002	0	2.3†
A: K=5, R=2, W=200	0	1	0	264 784 815	0	17 163†	56 022	0	22†
A: K=9, R=1, W=200	0	1	0	118 362 706	0	8 916†	78 730	0	42†
A: K=2, R=9, W=200	0	1	0	–	–	–	3 141 634	0	2 543
A: K=13, R=1, W=200	0	1	0	–	–	–	6 377 290	0	5 564†
C: K=4, R=2, W=1 000	0	200	0	6 315 296	0	364	8 002	0	2.3
D: K=3, R=3, W=1 000	0	2 500	0	20 391 072	0	1 109	20 184 036	0	10 283†
PAR (K messages, R retries, W)									
A: K=5, R=2, W=2	0	1	0	69 529 579	3 109 929	4 379†	1 955 414	60 702	840
A: K=5, R=2, W=3	0	1	0	69 529 579	3 109 929	4 379†	56 022	0	21
A: K=5, R=2, W=200	0	1	0	69 529 579	3 109 929	4 379†	56 022	0	22
A: K=4, R=3, W=200	0	1	0	308 419 431	6 150 740	19 416†	159 106	0	67†
A: K=8, R=1, W=200	0	1	0	31 925 127	2 789 814	2 202	26 242	0	11†
C: K=6, R=1, W=400	0	200	0	496 447	35 874	27	230 836	16 999	93
C: K=6, R=1, W=401	0	200	0	496 447	35 874	27	2 914	0	0.8
D: K=6, R=1, W=5 000	0	2 500	0	496 447	35 874	27	230 836	16 999	93
D: K=6, R=1, W=5 001	0	2 500	0	496 447	35 874	27	2 914	0	0.8
C: K=8, R=1, W=200	0	200	0	31 925 127	2 789 814	605$^{\ddagger 16}$	26 183 889	2 338 442	2 178$^{\ddagger 16}$
C: K=8, R=1, W=1 000	0	200	0	31 925 127	2 789 814	2 202	26 242	0	11
A: K=2, R=9, W=200	0	1	0	–	–	–	3 141 634	0	2 544
A: K=2, R=10, W=200	0	1	0	–	–	–	12 574 722	0	11 962†
A: K=2, R=11, W=200	0	1	0	–	–	–	50 315 266	0	4 466$^{\ddagger 16}$
A: K=13, R=1, W=200	0	1	0	–	–	–	6 377 290	0	5 498†
A: K=14, R=1, W=200	0	1	0	–	–	–	19 131 874	0	1 451$^{\ddagger 16}$
A: K=15, R=1, W=200	0	1	0	–	–	–	57 395 626	0	4 238$^{\ddagger 16}$
D: K=8, R=1, W=1 000	0	2 500	0	31 925 127	2 789 814	605$^{\ddagger 16}$	31 925 127	2 789 814	5 609$^{\ddagger 16}$
BRP (n chunks, MAX, T_1, TR, property)									
A: n=3, MAX=2, T_1=1 000, TR=4 003, (2)	0.5	1	0	3 311 730	92 574	182	7 968	0	4.2
A: n=3, MAX=2, T_1=1 000, TR=4 003, (4)	0.5	1	0	17 761 263	6 908 822	1 271	12 191	2 094	8.4
A: n=5, MAX=1, T_1=200, TR=403, (2)	0.5	1	0	29 053 689	959 143	1 966	47 749	0	35
A: n=6, MAX=1, T_1=200, TR=403, (2)	0.5	1	0	1 260 206 952	39 940 668	8 172$^{\ddagger 16}$	398 145	0	24$^{\ddagger 16}$
A: n=7, MAX=1, T_1=200, TR=403, (2)	0.5	1	0	–	–	–	3 317 887	0	4 257†
A: n=8, MAX=1, T_1=200, TR=403, (2)	0.5	1	0	–	–	–	27 646 841	0	2 704$^{\ddagger 16}$
A: n=2, MAX=5, T_1=200, TR=2 003, (2)	0.5	1	0	206 967 587	1 446 477	2 099$^{\ddagger 8}$	34 006	0	3.3$^{\ddagger 8}$
A: n=2, MAX=8, T_1=200, TR=3 203, (2)	0.5	1	0	–	–	–	2 221 702	0	3 060†

Configuration	Timing			Must			Mustard		
	L	U	sd	Exec.	Viol.	Time	Exec.	Viol.	Time
A: n=2, MAX=9, T_1=200, TR=3 603, (2)	0.5	1	0	–	–	–	8 899 846	0	14 342†
C: n=3, MAX=2, T_1=200, TR=1 400, (2)	0.5	200	0	3 311 730	92 574	182	931 860	0	482
C: n=3, MAX=2, T_1=200, TR=1 400, (4)	0.5	200	0	17 761 263	6 908 822	1 271	1 312 615	197 460	1 154
D: n=2, MAX=5, T_1=1 000, TR=17 500, (2)	0.5	2 500	0	206 967 587	1 446 477	1 674$^{\ddagger 16}$	25 088 242	0	3 224$^{\ddagger 16}$
C: n=2, MAX=1, T_1=1 000, TR=2 600, (2)	0.5	200	0	–	–	$>$ 11 285	3 244 777	0	3 310
Online shop (C clients, N, R, S, D)									
A: C=2, N=2, R=4, S=20, D=29 000	0	1	0	1 038 352	1 020 432	141	709 432	691 512	580
B: C=3, N=3, R=2, S=20, D=29 000	0	65	0	1 564 672	1 558 912	212	861 328	855 568	561
E: C=3, N=3, R=2, S=20, D=29 000	0	90 000	0	1 564 672	1 558 912	207	1 564 672	1 558 912	858
YARN (K AMs, horizon H, property)									
A-C: K=2, H=9, miss	0	0	0	31 933 996	45 689	3 164	14 385	372	11
D: K=2, H=9, miss	0	3	0	31 933 996	45 689	3 164	87 508	5 077	64
A-C: K=2, H=10, late-report	0	0	0	160 091 241	496 450	2 508$^{\ddagger 8}$	29 515	0	3.8$^{\ddagger 8}$
A-C: K=2, H=10, miss	0	0	0	159 156 296	223 993	17 050†	28 363	732	23
A-C: K=2, H=10, preempt	0	0	0	159 517 787	158 922	3 140$^{\ddagger 7}$	29 043	160	3.5$^{\ddagger 7}$
A-C: K=2, H=11, miss	0	0	0	793 245 760	1 108 529	8 185$^{\ddagger 16}$	55 555	1 448	3.8$^{\ddagger 16}$
D: K=2, H=11, miss	0	3	0	793 245 760	1 108 529	8 185$^{\ddagger 16}$	597 360	30 349	41$^{\ddagger 16}$
D: K=2, H=15, miss	0	3	0	–	–	–	27 585 108	1 411 133	2 758$^{\ddagger 16}$
A-C: K=2, H=17, miss	0	0	0	–	–	–	3 081 303	90 344	5 847†
A-C: K=2, H=18, miss	0	0	0	–	–	–	6 004 023	175 428	839$^{\ddagger 16}$
A-C: K=2, H=19, miss	0	0	0	–	–	–	11 700 581	343 368	1 742$^{\ddagger 16}$
A-C: K=4, H=5, miss	0	0	0	–	–	$>$ 10 007	1 988 993	0	1 941†
A-C: K=4, H=6, miss	0	0	0	–	–	–	14 731 889	0	17 537†
A-C: K=1, H=24, preempt	0	0	0	58 047 984	1 286 129	7 389†	6 391	385	11
A-C: K=1, H=48, preempt	0	0	0	–	–	–	1 072 159	66 997	7 466†
A-C: K=1, H=52, preempt	0	0	0	–	–	–	2 506 757	156 681	1 342$^{\ddagger 16}$